The pair singular series of a polynomial IV: the second spectrum

Jasper Reichardt*

Working draft of September 10, 2026

Abstract

Part III reduced the Cesàro form of the off-diagonal of the pair singular series of an irreducible quadratic f to "pieces" $\mathcal{P}_u(Y)$, one for each way the two roots of f can agree on a part u of the modulus, and showed that everything except the pieces' "windows" is understood. This paper is about what the pieces actually are. Numerically each piece is $O(\sqrt{Y})$, and $\mathcal{P}_u(Y)/\sqrt{Y}$ oscillates in $\log Y$ at the Laplace eigenvalues of the *even* Maass cusp forms of $\mathrm{SL}_2(\mathbb{Z})$, with amplitudes governed by the values of the forms at the Heegner points, or their integrals over the closed geodesics, of discriminant D: a second spectrum, next to the zeros of ζ_K that govern the diagonal (part II). We explain this by an explicit formula: the piece is a twisted Walfisz sum whose Dirichlet series is a sum over frequencies of Dirichlet series of Salié sums, which the Kuznetsov formula of half-integral weight continues to the half-line with poles at $\frac{1}{2} \pm it_j$, and the Katok–Sarnak correspondence identifies the residues as periods of Maass forms. We prove the formula for a single piece in Riesz-mean form [[target]], discuss the uniformity in u that the Cesàro conjecture of part I needs, and report the numerical evidence.

1 Introduction

In plain terms. Parts I and II showed that the main part of the fluctuation of prime values of a polynomial is governed by the zeros of a Dedekind zeta function. The remaining part, the "off-diagonal" of part III, turns out to be governed by a second, entirely different spectrum: the Laplace eigenvalues of the modular surface, the same numbers that govern the distribution of the roots of $x^2 \equiv D \pmod{d}$ (Hooley, Bykovskiĭ, Duke–Friedlander–Iwaniec). Only half of them appear, those of the even forms, and the size of each contribution is the value of the corresponding eigenfunction at a special point, or its integral along a special closed curve, determined by the discriminant of the polynomial. We found this in the data before we had the theory; the theory, once written down, predicts exactly the pattern of presence and absence that the data show.

Set-up. Notation as in part III: $f = t^2 + bt + c$, $D = b^2 - 4c$, $\lambda(p) = p/(p - 2\omega(p))$; for squarefree u with $\omega(u) \geq 1$ and $Q_u(x) = u^2x^2 - D$,

$$\mathcal{P}_u(Y) = \sum_{h \leq Y} (Y - h)\big(F_u(Q_u(h)) - \mathbb{E}F_u\big) - \tfrac{1}{2}(\mathbb{E}F_u - 1)Y, \qquad F_u(n) = \sum_{\substack{d \mid n,\ d>1 \\ d \text{ admissible}}} \frac{\lambda(d)}{d}, \qquad \mathbb{E}F_u = \sum_{\substack{d>1 \\ d \text{ adm}}} \frac{\lambda(d)\omega(d)}{d^2},$$

admissible meaning squarefree, all primes split, coprime to $2Du$. Part III, Theorem A′: $\mathrm{Off}^*_f(H) = c_{\mathrm{off}}(f)H + \sum_{u \leq H^{2/3+\varepsilon}} w(u)\mathcal{W}_u(H/u; \log H) + O(H(\log H)^{1-c})$, and the Cesàro form of Conjecture 1 of part I is equivalent to the windows being $o(H \log H)$.

*Independent researcher. Correspondence: jasperreichardt@gmail.com. This work was carried out with heavy use of a large language model; see the disclosure at the end.

Results.

- *The spectral formula for a piece* (Theorem 11, [to be proved]): for fixed f and u, and Riesz means of order $m \ge m_0$,

$$\mathcal{P}_u^{(m)}(Y) = c_u Y + Y^{1/2} \sum_j \left(\alpha_j(u,f) Y^{it_j} + \overline{\alpha_j(u,f)} Y^{-it_j}\right) \gamma_m(t_j) + O(Y^{1/2-\delta}),$$

  the sum over the even Maass cusp forms u_j of $\mathrm{SL}_2(\mathbb{Z})$ with Laplace eigenvalue $\frac{1}{4} + t_j^2$ (and, for $u > 1$, the newforms of the levels dividing u^2), with $\alpha_j(u,f)$ an explicit multiple of the Katok–Sarnak period of u_j at discriminant D, and $\gamma_m(t) \ll |t|^{-m-1}$.

- *Numerical confirmation* (§6): for five quadratics the even parameters explain the variance of $\mathcal{P}_u(Y)/\sqrt{Y}$ at the 92nd–100th percentile among random frequency sets, the odd ones at the 12th–66th; the dominant line $t_1 = 13.7798$ has stable amplitude and phase over two decades of $\log Y$; the two polynomials with exponentially small periods ($D = -8, -163$) show nothing.

- *What the Cesàro conjecture needs* (§7): the sum of the pieces over $u \le H^{2/3}$ converges to $c_{\mathrm{off}} H + o(H)$ provided the spectral formula holds with an error $O((H/u)^{1/2+\varepsilon} u^A)$, $A < \frac{1}{4}$; the level dependence of the Kuznetsov formula is therefore the whole remaining problem.

2 The piece as a twisted Walfisz sum

Throughout, f is an irreducible quadratic of discriminant D and u is squarefree with $\omega(u) \ge 1$. Call a modulus d *admissible* (for u) if it is squarefree and coprime to $2Du$. Only such d occur: if $p \mid Q_u(h) = u^2h^2 - D$ and $p \nmid 2Du$ then $D \equiv (uh)^2 \pmod{p}$, so $\left(\frac{D}{p}\right) = 1$ and p is split. The condition "every prime dividing d is split", which appears in part III, is therefore automatic here, and we do not impose it. For admissible d put

$$R_d = R_d^{(u)} = \{x \bmod d : \; u^2x^2 \equiv D\}, \qquad \rho(d) = |R_d| = \omega(d) = 2^{\nu(d)}, \qquad \rho_k(d) = \sum_{x \in R_d} \mathrm{e}\Big(\frac{kx}{d}\Big),$$

ν the number of prime factors and $\mathrm{e}(y) = e^{2\pi i y}$. Two properties are used constantly: R_d is *symmetric*, $x \in R_d \Rightarrow -x \in R_d$, and $0 \notin R_d$, both because $(D,d) = 1$; consequently every $\rho_k(d)$ is real. Write $\lambda(p) = p/(p-4)$ on split primes, λ multiplicative, and

$$F_u(n) = \sum_{\substack{d \mid n,\ d>1 \\ d \text{ admissible}}} \frac{\lambda(d)}{d}, \qquad \mathbb{E}F_u = \sum_{\substack{d>1 \\ d \text{ admissible}}} \frac{\lambda(d)\rho(d)}{d^2}, \qquad Q_u(h) = u^2h^2 - D, \tag{1}$$

both sums absolutely convergent ($\lambda(d)\rho(d) \ll d^{\varepsilon}$). *All sums over moduli below run over admissible $d > 1$*: including $d = 1$ would add 1 to F_u and to $\mathbb{E}F_u$, leaving (2) unchanged but spoiling the symmetry used in Lemma 1, since $R_1 = \{0\}$ is not a union of pairs $\{x, -x\}$ with $x \ne -x$. The *sharp piece* and the *Riesz pieces* are

$$S_u(t) = \sum_{h \le t} \big(F_u(Q_u(h)) - \mathbb{E}F_u\big), \qquad \mathcal{P}_u^{(m)}(Y) = \frac{1}{m!} \sum_{h \le Y} (Y-h)^m \big(F_u(Q_u(h)) - \mathbb{E}F_u\big) \quad (m \ge 0), \tag{2}$$

so that $\mathcal{P}_u^{(0)} = S_u$ and, for integers Y, $\mathcal{P}_u^{(1)}(Y) = \sum_{t<Y} S_u(t)$; up to the explicit constant of part III, Remark **??**, $\mathcal{P}_u^{(1)}(Y)$ is the piece $\mathcal{P}_u(Y)$ of part III. The connection with part III is that $F_u(Q_u(h))$ is the divisor-sum form of the inner sum of a piece: $d \mid Q_u(h)$ exactly when h lies in one of the $\rho(d)$ classes R_d.

Lemma 1 (sawtooth form)**.** *Let $\psi(y) = \frac{1}{2} - \{y\}$. For every $t \geq 0$,*

$$S_u(t) = \sum_{d \text{ admissible}} \frac{\lambda(d)}{d} \sum_{x \in R_d} \psi\Big(\frac{t-x}{d}\Big), \tag{3}$$

the series converging absolutely.

Proof. The right-hand side is not absolutely convergent term by term, since $\sum_d \lambda(d)\rho(d)/d$ diverges; it is summed with the terms x and $-x$ grouped, and converges in that sense, as shown at the end. Expanding F_u and counting,

$$\sum_{h \leq t} F_u(Q_u(h)) = \sum_d \frac{\lambda(d)}{d} \#\{h \leq t : \ d \mid Q_u(h)\} = \sum_d \frac{\lambda(d)}{d} \sum_{x \in R_d} \#\{h \leq t : \ h \equiv x \ (d)\},$$

the interchange being legitimate because $\sum_{d \mid Q_u(h)} \lambda(d)/d$ converges absolutely for each $h \leq t$ and there are finitely many h. Represent each $x \in R_d$ by $\langle x \rangle_d \in [1, d]$. For $0 \leq t$ and $1 \leq x \leq d$,

$$\#\{1 \leq h \leq t : \ h \equiv x \ (d)\} = \Big\lfloor \frac{t-x}{d} \Big\rfloor + 1,$$

valid also when $x > t$, where both sides vanish because $-1 < (t-x)/d < 0$. Writing $\lfloor y \rfloor = y - \frac{1}{2} + \psi(y)$,

$$\#\{h \leq t : \ h \equiv x \ (d)\} = \frac{t}{d} + \Big(\frac{1}{2} - \frac{x}{d}\Big) + \psi\Big(\frac{t-x}{d}\Big).$$

Now sum over $x \in R_d$. Since R_d is symmetric and $0 \notin R_d$, its representatives split into pairs $\{\langle x \rangle_d, \ d - \langle x \rangle_d\}$, and $(\frac{1}{2} - \frac{x}{d}) + (\frac{1}{2} - \frac{d-x}{d}) = 0$; hence $\sum_{x \in R_d} (\frac{1}{2} - \langle x \rangle_d / d) = 0$ for every admissible $d > 1$. Therefore

$$\sum_{h \leq t} F_u(Q_u(h)) = t \sum_d \frac{\lambda(d)\rho(d)}{d^2} + \sum_d \frac{\lambda(d)}{d} \sum_{x \in R_d} \psi\Big(\frac{t-x}{d}\Big) = t\, \mathbb{E}F_u + \sum_d \frac{\lambda(d)}{d} \sum_{x \in R_d} \psi\Big(\frac{t-x}{d}\Big),$$

which is (3). For the convergence, group the terms of a pair $\{x, -x\}$ and write $a = \langle x \rangle_d$ with $a \leq d/2$, so that the pair is $\{a, d-a\}$. If $d > 2t$ and $a > t$ then $\{(t-a)/d\} = (t - a + d)/d$ and $\{(t+a)/d\} = (t+a)/d$, so by the periodicity of ψ the two sawtooths sum to $-2t/d$; if $a \leq t$ their sum is bounded by 1, and this happens for at most $\rho(d)$ values with $a \leq t < d/2$, a set that is empty once $d > 2t$. Hence for $d > 2t$ each pair contributes $\ll t/d$ and

$$\sum_{d > 2t} \frac{\lambda(d)}{d} \cdot \frac{t\,\rho(d)}{d} \ \ll \ t \sum_{d > 2t} \frac{\lambda(d)\rho(d)}{d^2} \ \ll \ t^{\varepsilon},$$

while the range $d \leq 2t$ is a finite sum. So the paired series converges and (3) holds. □

In words. A piece counts, for each admissible modulus d, how far the number of $h \leq t$ with $d \mid Q_u(h)$ departs from its expectation, weighted by $\lambda(d)/d$; and that departure is a sawtooth evaluated at the roots. The linear term that normally accompanies such a count cancels because the roots come in pairs $\pm x$. This is the same cancellation that makes the diagonal of part I a sum of sawtooths $\{H/d\} - \frac{1}{2}$, and it is why the piece is a Walfisz-type sum: in the classical case $\sum_{n \leq x} n^{-1}(\frac{1}{2} - \{x/n\})$ one sums a sawtooth over all moduli with weight $1/n$, and here one sums it over admissible moduli, at the roots of a quadratic congruence, with weight $\lambda(d)/d$.

Lemma 2 (Weyl-sum form)**.** *For every $t \geq 0$ and every $K \geq 1$,*

$$S_u(t) = \frac{1}{\pi} \sum_{k \geq 1} \frac{1}{k} \sum_{d \text{ admissible}} \frac{\lambda(d)\rho_k(d)}{d} \sin\frac{2\pi k t}{d}, \tag{4}$$

the k-sum converging boundedly. Consequently, for the Riesz pieces of order $m \ge 1$ and integer Y,

$$\mathcal{P}_u^{(m)}(Y) = \frac{1}{\pi}\sum_{k\ge1}\frac{1}{k}\sum_{d}\frac{\lambda(d)\rho_k(d)}{d}\,\Phi_m\Big(\frac{2\pi k}{d};Y\Big), \qquad \Phi_m(\theta;Y) = \frac{1}{m!}\sum_{h\le Y}(Y-h)^m\sin(\theta h), \tag{5}$$

with $\Phi_m(\theta;Y) \ll_m \min\big(Y^{m+1},\ Y^m\|\theta/2\pi\|^{-1}\big)$, $\|\cdot\|$ the distance to $\mathbb{Z}$.

Proof. The Fourier expansion $\psi(y) = \sum_{k\ge1}\sin(2\pi ky)/(\pi k)$ holds for $y \notin \mathbb{Z}$, boundedly and uniformly on compact subsets of $\mathbb{R}\setminus\mathbb{Z}$, and the partial sums are uniformly bounded. Insert it into (3) and use, for each d,

$$\sum_{x\in R_d}\sin\frac{2\pi k(t-x)}{d} = \Im\Big[\mathrm{e}\Big(\frac{kt}{d}\Big)\overline{\rho_k(d)}\Big] = \rho_k(d)\,\sin\frac{2\pi kt}{d},$$

since $\rho_k(d)$ is real. The interchange of the k-sum with the d-sum is justified by the uniform boundedness of the partial sums of the Fourier series together with the absolute convergence established in Lemma 1 (dominated convergence for series). For (5), apply $\frac{1}{m!}\sum_{h\le Y}(Y-h)^m$ to $t=h$ in (4), which is a finite sum, and evaluate Φ_m by summation by parts m times against the geometric sum $\sum_{h\le Y}\mathrm{e}(\theta h/2\pi) \ll \min(Y, \|\theta/2\pi\|^{-1})$. ☐

Remark 3. Two features of (5) decide everything that follows. First, the frequencies k enter only through the *Weyl sums* $\rho_k(d)$ of the dilated roots, so the arithmetic of the problem is entirely contained in the Dirichlet series

$$W_k(s) = \sum_{d\ \text{admissible}}\frac{\lambda(d)\rho_k(d)}{d^s} \qquad (\Re s > 1), \tag{6}$$

one for each k. Second, the weight $1/k$ in front decays only like k^{-1}, so a bound for W_k that loses a power of k is expensive: this is the same phenomenon that forced the condition $\theta + 6B < 1$ in part III, Theorem A, and it is the reason we do not bound the $\rho_k(d)$ one frequency at a time. The route of this paper is instead to continue $W_k(s)$ analytically, where the dependence on k appears as a Fourier coefficient of an automorphic form and therefore *decays* on average.

Proposition 4 (the Mellin representation)**.** *Let $m \ge 1$ and let*

$$A_u(s) = \sum_{h\ge1}\frac{F_u(Q_u(h)) - \mathbb{E}F_u}{h^s} \qquad (\Re s > 1).$$

Then A_u extends to a function holomorphic in $\Re s > \frac{1}{2}$ except possibly for the singularities inherited from the W_k, and for integer $Y \ge 2$ and any $c > 1$,

$$\mathcal{P}_u^{(m)}(Y) = \frac{1}{2\pi i}\int_{(c)} A_u(s)\,\frac{Y^{s+m}}{s(s+1)\cdots(s+m)}\,ds. \tag{7}$$

Moreover, for $0 < \Re s < 1$,

$$A_u(s) = \frac{2\Gamma(1-s)}{(2\pi)^{1-s}}\,\sin\frac{\pi s}{2}\,\sum_{k\ge1}k^{s-1}\,W_k(s+1) \ + \ \big(\mathit{entire}\big), \tag{8}$$

whenever the k-sum converges, where W_k is as in (6).

Proof. (7) is the Riesz–Perron formula, valid because $F_u(Q_u(h)) - \mathbb{E}F_u \ll h^\varepsilon$ so that A_u converges absolutely for $\Re s > 1$. For (8), expand F_u and sum over each residue class:

$$\sum_{h \geq 1} \frac{F_u(Q_u(h))}{h^s} = \sum_d \frac{\lambda(d)}{d^{1+s}} \sum_{x \in R_d} \zeta\Big(s, \frac{\langle x \rangle_d}{d}\Big),$$

$\zeta(s, \alpha)$ the Hurwitz zeta function. Since $\zeta(s, \alpha)$ has residue 1 at $s = 1$ for every α, the pole of the d-th term is $\rho(d)\lambda(d)d^{-1-s}$, and summing over d these residues total $\mathbb{E}F_u$, which is exactly the pole of $\mathbb{E}F_u\zeta(s)$: each bracket $\lambda(d)d^{-1-s}\sum_x \zeta(s, \langle x \rangle_d/d) - \lambda(d)\rho(d)d^{-2}\zeta(s)$ is entire, which is the statement that A_u continues past $s = 1$. Hurwitz's formula,

$$\zeta(s, \alpha) = \frac{2\Gamma(1-s)}{(2\pi)^{1-s}}\Big[\sin\frac{\pi s}{2}\sum_{k \geq 1}\frac{\cos 2\pi k\alpha}{k^{1-s}} + \cos\frac{\pi s}{2}\sum_{k \geq 1}\frac{\sin 2\pi k\alpha}{k^{1-s}}\Big],$$

valid for $\Re s < 0$ and by continuation for $\Re s < 1$ with the series interpreted as the analytic continuations of the periodic zeta functions, gives after summing over the symmetric set R_d

$$\sum_{x \in R_d} \zeta\Big(s, \frac{\langle x \rangle_d}{d}\Big) = \frac{2\Gamma(1-s)}{(2\pi)^{1-s}}\sin\frac{\pi s}{2}\sum_{k \geq 1}\frac{\rho_k(d)}{k^{1-s}},$$

the sine term vanishing identically because $\sum_{x \in R_d} \sin(2\pi kx/d) = 0$ by symmetry. Multiplying by $\lambda(d)d^{-1-s}$ and summing over d gives (8). □

In words. The piece is determined by one Dirichlet series for each frequency, namely the Dirichlet series of the Weyl sums of the roots of $u^2x^2 \equiv D$; the symmetry of the root set kills the odd half of Hurwitz's formula, so only the even half survives. That the odd half disappears here is the first appearance of the parity phenomenon that will select the even Maass forms in §4, and it has the same source: the root set of a quadratic congruence is invariant under $x \mapsto -x$.

3 Salié sums and the Dirichlet series of the Weyl sums

By Remark 3 everything now depends on the series $W_k(s) = \sum_d \lambda(d)\rho_k(d)d^{-s}$. This section identifies its coefficients as Kloosterman sums of half-integral weight, which is what makes an analytic continuation possible. Write, for $c \geq 1$ odd with $(D, c) = 1$ and $h \in \mathbb{Z}$,

$$\mathcal{W}_h(D; c) = \sum_{b \bmod c,\ b^2 \equiv D} \mathrm{e}\Big(\frac{hb}{c}\Big), \tag{9}$$

the Weyl sum of the roots of the *fixed* quadratic congruence, and let $K_\chi(m, n; p) = \sum^*_{x \bmod p}\big(\frac{x}{p}\big)\mathrm{e}\big(\frac{mx + n\bar{x}}{p}\big)$ be the Kloosterman sum twisted by the Legendre symbol, $\varepsilon_p = 1$ or i according as $p \equiv 1$ or 3 (mod 4).

Lemma 5 (the dilation is a change of frequency)**.** *For admissible d and $k \geq 1$, with $\bar{u}$ the inverse of u modulo d,*

$$\rho_k(d) = \mathcal{W}_{k\bar{u}}(D; d).$$

Proof. $x \mapsto b = ux$ is a bijection of $R_d^{(u)} = \{x : u^2x^2 \equiv D\}$ onto $\{b : b^2 \equiv D\}$ because $(u, d) = 1$, and $kx \equiv k\bar{u}b \pmod{d}$. □

So the dilation by u does not change the quadratic; it moves the frequency from k to $k\bar{u}$, a frequency that depends on the modulus. This is the exact point at which the pieces with $u > 1$ leave the range of the theorems of Duke, Friedlander and Iwaniec, which are stated for frequencies fixed independently of the modulus.

Lemma 6 (Salié)**.** *Let p be an odd prime with $\left(\frac{D}{p}\right) = 1$ and let $h \in \mathbb{Z}$ with $p \nmid h$. Then*

$$\mathcal{W}_{2h}(D;p) = \frac{1}{\varepsilon_p \sqrt{p}}\, K_\chi(D, h^2; p). \tag{10}$$

For odd squarefree $c = c_1c_2$ with $(c_1, c_2) = 1$ and $(D, c) = 1$,

$$\mathcal{W}_h(D;c) = \mathcal{W}_{h\overline{c_2}}(D;c_1)\, \mathcal{W}_{h\overline{c_1}}(D;c_2), \tag{11}$$

so (10) determines $\mathcal{W}_h(D;c)$ for every odd squarefree c coprime to $2D$.

Proof. (11) is the Chinese remainder theorem: $b \mapsto (b \bmod c_1, b \bmod c_2)$ is a bijection of the roots modulo c onto the pairs of roots, and $b/c \equiv b_1\overline{c_2}/c_1 + b_2\overline{c_1}/c_2 \pmod 1$. For (10), substitute $b = 2\bar{h}y$ in (9), which is a bijection of the roots of $b^2 \equiv D$ onto the roots of $y^2 \equiv h^2 D\bar{4}$ and turns $\mathrm{e}(2hb/p)$ into $\mathrm{e}(2y/p)$; hence $\mathcal{W}_{2h}(D;p) = \sum_{y^2 \equiv h^2 D\bar{4}} \mathrm{e}(2y/p)$. Salié's evaluation of the twisted Kloosterman sum, $K_\chi(m,n;p) = \varepsilon_p\sqrt{p}\left(\frac{m}{p}\right)\sum_{y^2\equiv mn\,(p)} \mathrm{e}\left(\frac{2y}{p}\right)$ [3, §4.5], applied with $m = D$, $n = h^2$ gives (10), the factor $\left(\frac{D}{p}\right)$ being 1 because p is split. [Both identities were verified numerically for all split $p \le 37$, all $D \in \{-4, -3, 5, 8, 12\}$ and $h \in \{1, 2, 3, 5\}$, and (11) for composite c; the reference for Salié's evaluation should be pinned to a precise statement of the normalisation.] ☐

Remark 7 (what has been achieved)*.* Combining Lemmas 5 and 6, for even k and admissible d the coefficient of $W_k(s)$ is, up to the elementary factor $\varepsilon_d^{-1}d^{-1/2}$, a Kloosterman sum of half-integral weight with the two arguments D and $(k\bar{u}/2)^2$. The first argument is fixed by the polynomial; the second is a square, and it is where the frequency and the dilation sit. This is the shape to which the Kuznetsov formula of weight one half applies, and it explains the two features of the numerical law of §6: the *eigenvalues* that appear are those of the automorphic spectrum of weight one half, hence, through the Shimura correspondence, those of the Maass forms of $\mathrm{SL}_2(\mathbb{Z})$; and the *amplitudes* are products of two Fourier coefficients, one at D and one at a square, the first being the Katok–Sarnak period of §4 and the second, by Shimura's relation for coefficients at squares, a Hecke eigenvalue of the lift.

The continuation of W_k. What is needed is the following, which we do not prove here.

Hypothesis 8 (K)**.** *For each $k \ge 1$ and each admissible class of moduli, $W_k(s)$ continues meromorphically to $\Re s > \frac{1}{2}$, with at most simple poles at $s = \frac{1}{2} \pm it_j$, $\frac{1}{4} + t_j^2$ running over the Laplace eigenvalues of the weight-$\frac{1}{2}$ spectrum of level $4u^2$, with residues*

$$\operatorname*{Res}_{s=\frac{1}{2}+it_j} W_k(s) = \gamma(t_j)\,\overline{\rho_j(|D|)}\,\rho_j\big((k/2)^2\big)\,(1 + o(1))$$

in the normalisation of [2], and with polynomial growth in $|\Im s|$ on vertical lines in $\Re s > \frac{1}{2} + \delta$.

For $u = 1$ and k fixed this is in substance the content of Bykovskiĭ's spectral treatment of Hooley's sums [1] and of [2, §§2–4]; the two features that are not in the literature are the uniformity in k, needed because (5) sums over k with weight $1/k$, and the uniformity in the level $4u^2$, needed because the Cesàro conjecture sums over u (§7). [Section to be completed: the precise form of the Kuznetsov formula for the theta multiplier (Proskurin), the admissibility sieve on the moduli (Möbius over ℓ^2 and the split condition, as in part III, Lemma 2.1), and the twist by $\lambda = 1 * \kappa$.]

4 Katok–Sarnak and the parity rule

Hypothesis K places the poles of W_k at $\frac{1}{2} \pm it_j$, t_j running over the weight-$\frac{1}{2}$ spectrum, with residues proportional to $\overline{\rho_j(|D|)}\rho_j((k/2)^2)$. Two classical inputs turn this into a statement about the Maass forms of $\mathrm{SL}_2(\mathbb{Z})$.

Shimura and Katok–Sarnak. The Shimura correspondence sends the weight-$\frac{1}{2}$ forms of level 4 in Kohnen's plus space to weight-0 Maass forms of level 1 with the same spectral parameter, and the image consists of the *even* forms; the odd Maass forms correspond to weight $\frac{3}{2}$. Katok and Sarnak [4] identify the Fourier coefficients of the weight-$\frac{1}{2}$ form at a fundamental discriminant D with a period of the corresponding Maass form u_j: for $D < 0$ the sum of u_j over the Heegner points of discriminant D, and for $D > 0$ the integral of u_j over the closed geodesics of discriminant D. Writing $\mathrm{Per}_D(u_j)$ for that period,

$$\rho_j(|D|) = \kappa(t_j, D)\,\mathrm{Per}_D(u_j) \tag{12}$$

with an explicit factor κ. [Pin κ and the normalisation of Per to [4]; only $|\kappa|$, and only its dependence on D, is used below.]

Proposition 9 (parity)**.** *If u_j is an odd Maass form then $\mathrm{Per}_D(u_j) = 0$, for every discriminant D. Consequently only the even Maass forms of $\mathrm{SL}_2(\mathbb{Z})$ can contribute to the pieces.*

Proof. Let $\iota(z) = -\bar{z}$, the reflection in the imaginary axis; u_j is odd exactly when $u_j \circ \iota = -u_j$. For $D < 0$ the Heegner points of discriminant D are the roots $z_Q \in \mathbb{H}$ of the integral binary quadratic forms $Q = [a, b, c]$ of discriminant D, one for each class, and $\iota(z_{[a,b,c]}) = z_{[a,-b,c]}$; since $[a, b, c] \mapsto [a, -b, c]$ is the inversion of the class group, it permutes the classes, so ι permutes the set of Heegner points. Hence $\mathrm{Per}_D(u_j) = \sum_Q u_j(z_Q) = \sum_Q u_j(\iota z_Q) = -\mathrm{Per}_D(u_j)$, which forces $\mathrm{Per}_D(u_j) = 0$. For $D > 0$ the same involution permutes the closed geodesics S_Q attached to the classes, and reverses none of the identifications used in defining the cycle integral, so the same computation applies to $\mathrm{Per}_D(u_j) = \sum_Q \int_{\Gamma_Q \backslash S_Q} u_j\, ds$. □

In words. The Heegner points of a given discriminant, and likewise the closed geodesics, are symmetric about the imaginary axis; an odd Maass form is antisymmetric about it; so its period vanishes. This is the third appearance of one and the same symmetry. In §2 it made the linear term of the sawtooth identity vanish; in Proposition 4 it killed the odd half of Hurwitz's formula; here it removes the odd half of the spectrum. All three are the statement that the roots of a quadratic congruence come in pairs $\pm x$.

Corollary 10 (the shape of a piece)**.** *Assume Hypothesis K. Then in the notation of §5 the spectral sum runs over the even Maass cusp forms of $\mathrm{SL}_2(\mathbb{Z})$ only, and the coefficient of the line t_j is proportional to $\mathrm{Per}_D(u_j)$.*

The size of the periods, and a visibility threshold. For $D < 0$ of class number one the period is the single value $u_1(z_D)$ at $z_D = (-b + i\sqrt{|D|})/2$. Through the Fourier expansion $u_j(z) = 2\sqrt{y} \sum_{n\geq 1} a_j(n) K_{it_j}(2\pi n y) \cos(2\pi n x)$ this is governed by $K_{it_j}(2\pi y)$ with $y = \sqrt{|D|}/2$. The Bessel function $K_{it}(x)$ of imaginary order is of size $e^{-\pi t/2}$ and oscillatory while $x < t$, and only begins to decay in x once $x > t$. Hence the periods of a fixed form u_j are all of comparable size for

$$|D| < \left(\frac{t_j}{\pi}\right)^2, \tag{13}$$

and collapse exponentially beyond it. For the first even form, $t_1 = 13.7798$, the threshold is $|D| \approx 19.2$. This is a falsifiable prediction: the first line should be visible for the small discriminants and invisible for $|D| \geq 43$, while the higher lines, having larger t_j, should persist further. Section 6 reports what is observed.

5 The spectral formula for a piece

Theorem 11 ([target])**.** *[statement as in the introduction, with the precise hypotheses on u and m_0.]*

6 Numerical evidence

All computations are exact rational or high-precision arithmetic; the scripts are in the repository (`research/paper-IV/scripts/`). For each f and u we compute $\mathcal{P}_u^{(1)}(Y)$ for all $Y \le 10^7$ (a sieve of $Q_u(h)$ for $h \le Y$, with $\mathbb{E}F_u$ evaluated to 22 significant digits, since an error δ in $\mathbb{E}F_u$ enters as $\delta Y^2/2$), sample it on a logarithmic grid of 4096 points in $\log Y \in [\log 10^3, \log Y_{\max}]$, remove the mean and the linear trend in $\log Y$, and regress the result on

$$\big\{\cos(t_j \log Y),\ \sin(t_j \log Y)\big\}_{j=1}^{6}$$

for the six smallest even spectral parameters, for the six smallest odd ones, and for 300 random sets of six frequencies drawn uniformly from $[12, 25]$. Reported is the fraction of variance explained, R^2, and where it falls among the random sets.

Table 1: Fit of $\mathcal{P}_u^{(1)}(Y)/\sqrt{Y}$ to the even and to the odd spectral parameters of $\mathrm{SL}_2(\mathbb{Z})$, against 300 random frequency sets of the same size ($Y \le 10^7$; $u = 1$ throughout except where shown).

f	D	R^2 even	percentile	R^2 odd	percentile	verdict
t^2-2	8	0.217	100	0.036	34	even
t^2+1	-4	0.152	100	0.028	20	even
t^2+t+1	-3	0.139	100	0.037	35	even
t^2+t+2	-7	0.089	100	0.051	76	even
t^2-3	12	0.066	99	0.037	66	even
t^2+t-1	5	0.062	92	0.036	32	even
t^2+t+5	-19	0.047	92	0.044	88	inconclusive
t^2+2	-8	0.031	47	0.054	86	none
t^2+t+3	-11	0.009	44	0.023	100	none

Table 1 is the evidence for Proposition 9. Six of the nine polynomials place the even set at the 92nd percentile or above, four of them above every one of the 300 random sets; the odd set never rises above the 88th percentile except once, at the 100th for t^2+t+3, where the even set is at the 44th and the fit is at the level of the noise. For the dominant line $t_1 = 13.7798$ the fitted amplitude and phase are stable when the range of $\log Y$ is cut in half: for t^2-2, amplitudes 0.019 and 0.018 with phases 0.89 and 0.61; for t^2+t+1, 0.021 and 0.017 with phases 1.01 and 1.03; for t^2+1 at $u = 2$, 0.042 and 0.042 with phases -0.30 and 0.07. A spurious fit does not reproduce its phase on an independent range.

The periods. Using the Fourier coefficients of the first even form (LMFDB label `1.0.1.3.1`, $t_1 = 13.7797513$) we evaluated u_1 at the Heegner points of the nine imaginary discriminants of class number one (`scripts/maass-period.py`).

Table 2: The Katok–Sarnak period of the first even Maass form at the Heegner point of discriminant $D < 0$, class number one, relative to $D = -4$; and the fitted amplitude of the line t_1, relative to $D = -4$.

D	-3	-4	-7	-8	-11	-19	-43	-67	-163
$\lvert u_1(z_D)\rvert/\lvert u_1(i)\rvert$	0.649	1	0.542	0.318	0.302	0.444	$6.6\cdot10^{-3}$	$1.0\cdot10^{-4}$	$2.2\cdot10^{-10}$
fitted amplitude ratio	0.483	1	0.111	0.281	0.461	1.358	—	—	—
signal present	yes	yes	yes	no	no	marginal	no	no	no

Table 2 confirms the mechanism at the level of orders of magnitude and no further. The threshold (13) sits at $|D| \approx 19.2$: below it every period lies within a factor 3 of every other, and every one of the six discriminants with $|D| \le 19$ has an amplitude within a factor 12 of the others; above it the periods fall by two, four and ten orders of magnitude at $D = -43, -67, -163$, and no signal is seen at any of them. In particular the absence at $D = -163$, the polynomial t^2+t+41, is explained. What the table does *not* confirm is the amplitude law itself: the two rows disagree by as much as a factor 5 at $D = -7$. Two unmodelled effects are of that size, the local factors, since $\mathbb{E}F_u$ ranges over $[0.35, 1.60]$ across these polynomials and enters multiplicatively, and the uncertainty of the fitted amplitudes themselves, which for $D = -7$ is a factor 2 between the two halves of the range. We record the amplitude law as a prediction to be tested, not as a confirmed fact.

The weight, and a model piece. The absences at $D = -8$ and $D = -11$, whose periods are of ordinary size, are not explained by anything above, and they are not isolated: among the fourteen polynomials we tested, the six showing no signal are exactly those for which the prime 3 splits in $\mathbb{Q}(\sqrt{D})$. That is a statement about the weight, not about the roots. Indeed $\lambda(3) = 3/(3-4) = -3$ is the only negative value taken by λ and the largest in modulus, and $|\lambda(3)\rho(3)/3| = 2 > 1$, so at $\Re s = \frac{1}{2}$ the local factor at 3 of the Euler product for W_k has modulus up to $3 \cdot 2 \cdot 3^{-1/2} = 3.46$: it is the one local factor that can dominate the spectral term. Removing the weight confirms this. Define the *model piece* by replacing λ by 1 in (1), so that F becomes the restricted divisor function $\sum_{d|n,\ d \text{ adm}} 1/d$ and $\mathcal{P}^{(1)}$ is a Riesz mean of σ_{-1} along Q_u. Repeating the regression:

Table 3: The model piece ($\lambda \equiv 1$) at $Y \leq 10^7$, $u = 1$: fit to the even and odd spectral parameters, with the weighted piece for comparison.

D	R^2 even	percentile	R^2 odd	percentile	weighted piece
8	0.319	100	0.036	14	0.217 (100)
−4	0.237	100	0.027	13	0.152 (100)
−3	0.139	100	0.019	13	0.139 (100)
−8	0.109	100	0.029	17	0.031 (47)
17	0.098	99.3	0.017	7	0.078 (99.3)
−11	0.078	100	0.015	11	0.009 (44)

All six are significant, every odd set falls below the 17th percentile, and the two discriminants that showed nothing under the weight show a clean signal without it. So the apparent rule at $p = 3$ is an artefact of λ; nothing arithmetic is hidden there. Two consequences matter for what follows. The phenomenon is *stronger* without the weight, so the model piece is the better object to measure; and its Dirichlet series is $\sum_d d^{-s}\mathcal{W}_k(D; d)$ with no multiplicative weight at all, a Salié zeta function carrying only a squarefree and a coprimality condition. That is the object we can hope to continue, and §5 is stated for it; the weighted piece, which is the one the off-diagonal of part III requires, is the model piece perturbed by $\lambda = 1 * \kappa$ with $\kappa(p) = 4/(p-4) = O(1/p)$.

The limits of the present computations, and what is planned. Every number reported above comes from a sieve of $Q_u(h)$ for $h \leq 10^7$, which runs in seconds to minutes on a laptop; the scripts are in the repository and each states its cost in its header. Three of the questions raised here are limited by that range rather than by anything conceptual, and are settled by longer runs of exactly the same code:

1. the amplitude law of Table 2, whose comparison is currently blunted by an uncertainty of up to a factor 2 in the fitted amplitudes; the uncertainty falls like $Y^{-1/2}$ in the length of the $\log Y$ range, so $Y = 10^9$ would roughly halve it and $Y = 10^{12}$ would settle the discrepancy at $D = -7$;

2. the resolution in the frequency, which is $2\pi/\log(Y_{\max}/Y_{\min}) \approx 0.68$ at present and therefore cannot separate the lines 19.42 and 19.48, the first even and the first odd parameter of

that size; this is the single measurement that would test the parity rule against an adjacent pair rather than against a whole set;

3. the selection rule at $p = 3$, where the sample is fourteen polynomials and should be a few hundred.

None of the three needs a new method, only more arithmetic, and we intend to carry the computation out on appropriate hardware and report it separately. We state them here so that a reader can see exactly which of our assertions rest on 10^7 and which do not: the parity rule of Table 1 and the threshold of Table 2 are stable across the halves of the present range and do not, the amplitude law does.

7 Uniformity in u and the Cesàro conjecture

[The condition $A < \frac{1}{4}$; what the spectral large sieve gives; the windows of part III in spectral terms.]

AI-assisted research: disclosure

This work was carried out with heavy use of large language models, the Anthropic models Claude Fable 5.1 and Claude Opus 5, used interactively by the author in September 2026 as research assistants. The author posed the questions, set the direction and the scope, and made every decision about what to pursue, claim and submit. The model was used to draft mathematical arguments, to write and run the software behind every number in the paper, to search the literature, and to draft the text; all of this was done in dialogue with the author, who verified the code and the numerical results, reviewed the proofs, and takes full responsibility for the content. The model is not an author. The complete record of the work is in the public repository `https://github.com/gewure/ulamnd` (directory `research/`).

References

[1] V. A. Bykovskiĭ. Spectral expansions of certain automorphic functions and their number-theoretic applications. *Zap. Nauchn. Sem. LOMI*, 134:15–33, 1984. English transl.: J. Soviet Math. 36 (1987), 8–21.

[2] W. Duke, J. B. Friedlander, and H. Iwaniec. Weyl sums for quadratic roots. *Int. Math. Res. Not. IMRN*, (11):2493–2549, 2012.

[3] Henryk Iwaniec. *Topics in Classical Automorphic Forms*, volume 17 of *Graduate Studies in Mathematics*. American Mathematical Society, 1997.

[4] Svetlana Katok and Peter Sarnak. Heegner points, cycles and Maass forms. *Israel Journal of Mathematics*, 84:193–227, 1993.